

Co-2 Assessment tool-2

Q1) Chart Selection:

1. Comparing the population of 8 countries in the current year

Chart Type: Bar Chart (Amounts)

Justification: A bar chart is the best choice because it clearly compares the population values of different countries. It makes it easy to identify the largest and smallest populations.

2. Showing how delivery times vary across 5 courier companies

Chart Type: Box Plot (Distribution)

Justification: A box plot shows the distribution of delivery times, including the median, spread, and any outliers. It is ideal for comparing variation among multiple courier companies.

3. Showing the proportion of a household budget spent on rent, food, transport, and savings

Chart Type: Pie Chart (Proportion)

Justification: A pie chart clearly displays how each expense contributes to the total household budget. It helps compare the percentage share of each category.

4. Showing the relationship between hours studied and exam marks for 50 students

Chart Type: Scatter Plot (x–y Relationship)

Justification: A scatter plot is best for showing the relationship between two numerical variables. It helps identify trends, patterns, or correlations between study hours and exam marks.

5. Showing average temperature for each month across 3 different cities at once

Chart Type: Line Chart (Temporal)

Justification: A line chart effectively shows changes in temperature over time. Using separate lines for each city makes it easy to compare monthly temperature trends.

6. Showing which airports are directly connected by flight routes

Chart Type: Network Diagram (Network)

Justification: A network diagram represents airports as nodes and flight routes as connecting edges. It clearly shows direct connections and the overall flight network.

Q2) R code:

```
# Load library
```

```
library(ggplot2)
```

```
# Create the data frame
```

```
app_downloads <- data.frame(  
  App_Category = c("Social Media", "Gaming",  
    "Productivity",  
      "Finance", "Health & Fitness",  
    "Education"),  
  Downloads_Millions = c(58, 46, 27, 19, 33, 24)  
)
```

```
# Sort data in ascending order
```

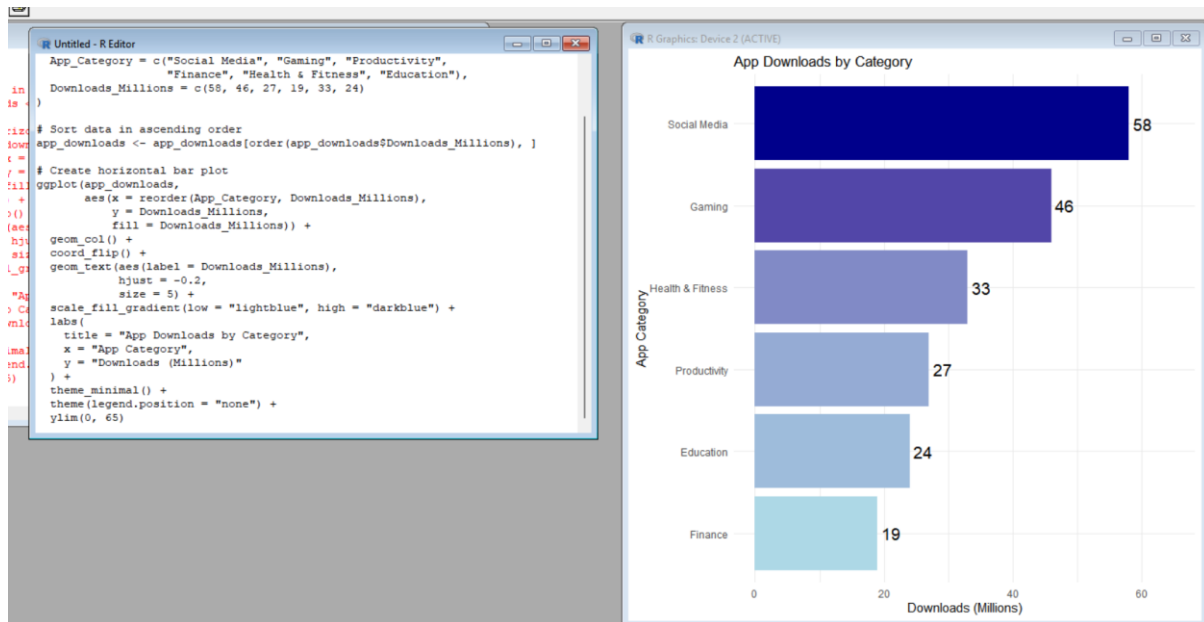
```
app_downloads <-  
app_downloads[order(app_downloads$Downloads_Mil  
lions), ]
```

```
# Create horizontal bar plot
```

```
ggplot(app_downloads,  
  aes(x = reorder(App_Category,  
    Downloads_Millions),  
    y = Downloads_Millions,
```

```
    fill = Downloads_Millions)) +  
geom_col() +  
coord_flip() +  
geom_text(aes(label = Downloads_Millions),  
          hjust = -0.2,  
          size = 5) +  
scale_fill_gradient(low = "lightblue", high =  
"darkblue") +  
labs(  
  title = "App Downloads by Category",  
  x = "App Category",  
  y = "Downloads (Millions)"  
) +  
theme_minimal() +  
theme(legend.position = "none") +  
ylim(0, 65)
```

Output:



Justification:

A **horizontal bar chart** is used because it clearly compares the number of downloads across different app categories. The horizontal layout makes long category names easier to read, while sorting the bars in ascending order helps quickly identify the categories with the lowest and highest downloads. The color gradient further emphasizes the differences in download counts, and the value labels provide the exact number of downloads for each category.

Q3) i) R code for grouped bar chart:

```
# Load library
```

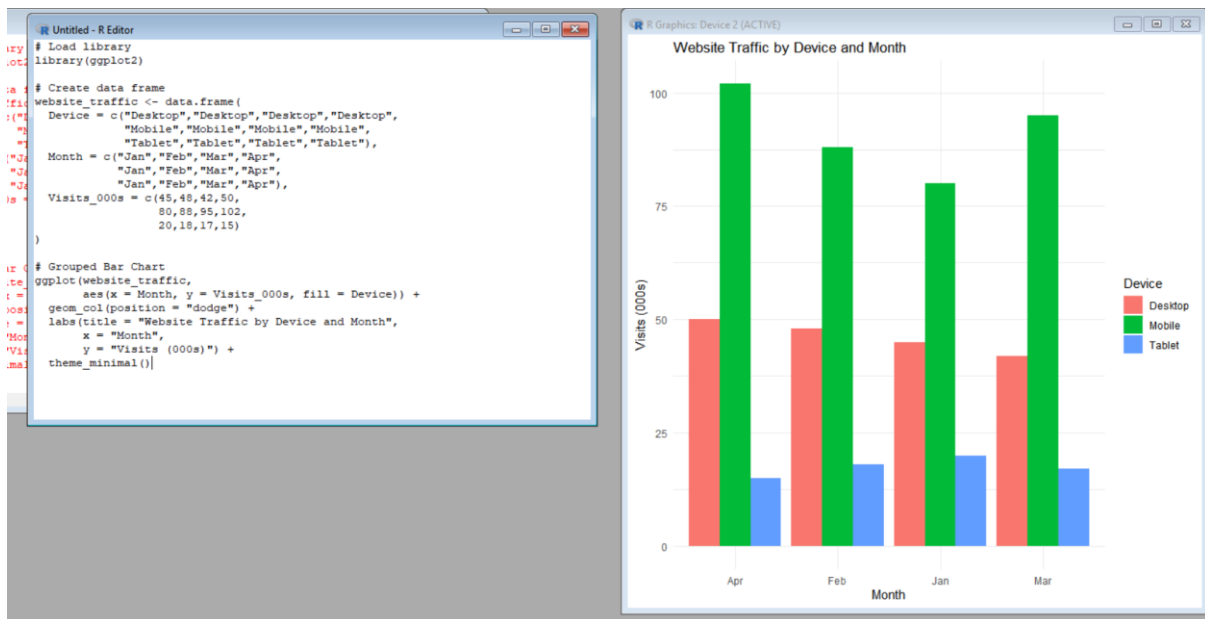
```
library(ggplot2)
```

```
# Create data frame
website_traffic <- data.frame(
  Device =
    c("Desktop","Desktop","Desktop","Desktop",
      "Mobile","Mobile","Mobile","Mobile",
      "Tablet","Tablet","Tablet","Tablet"),
  Month = c("Jan","Feb","Mar","Apr",
    "Jan","Feb","Mar","Apr",
    "Jan","Feb","Mar","Apr"),
  Visits_000s = c(45,48,42,50,
    80,88,95,102,
    20,18,17,15)
)
```

```
# Grouped Bar Chart
ggplot(website_traffic,
  aes(x = Month, y = Visits_000s, fill = Device)) +
  geom_col(position = "dodge") +
  labs(title = "Website Traffic by Device and Month",
```

```
x = "Month",
y = "Visits (000s)" +
theme_minimal()
```

Output:

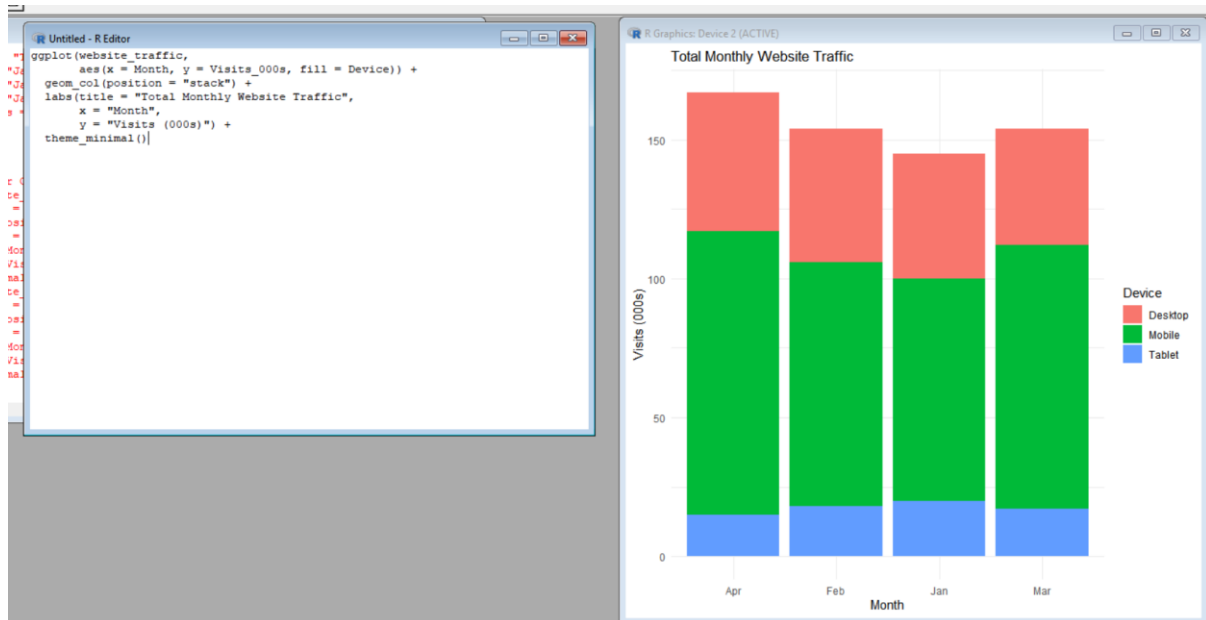


ii) R code for stacked bar chart:

```
ggplot(website_traffic,
  aes(x = Month, y = Visits_000s, fill = Device)) +
  geom_col(position = "stack") +
  labs(title = "Total Monthly Website Traffic",
        x = "Month",
        y = "Visits (000s)" +
```

theme_minimal()

Output:



iii) R code for dot plot:

Load library

library(ggplot2)

Create the data frame

website_traffic <- data.frame(

Device =

c("Desktop","Desktop","Desktop","Desktop",

"Mobile","Mobile","Mobile","Mobile",


```
      "Tablet","Tablet","Tablet","Tablet"),  
Month = c("Jan","Feb","Mar","Apr",  
          "Jan","Feb","Mar","Apr",  
          "Jan","Feb","Mar","Apr"),  
Visits_000s = c(45,48,42,50,  
                80,88,95,102,  
                20,18,17,15)  
)
```

```
# Calculate total visits for each device
```

```
device_total <- aggregate(Visits_000s ~ Device,  
                           data = website_traffic,  
                           FUN = sum)
```

```
# Rename column
```

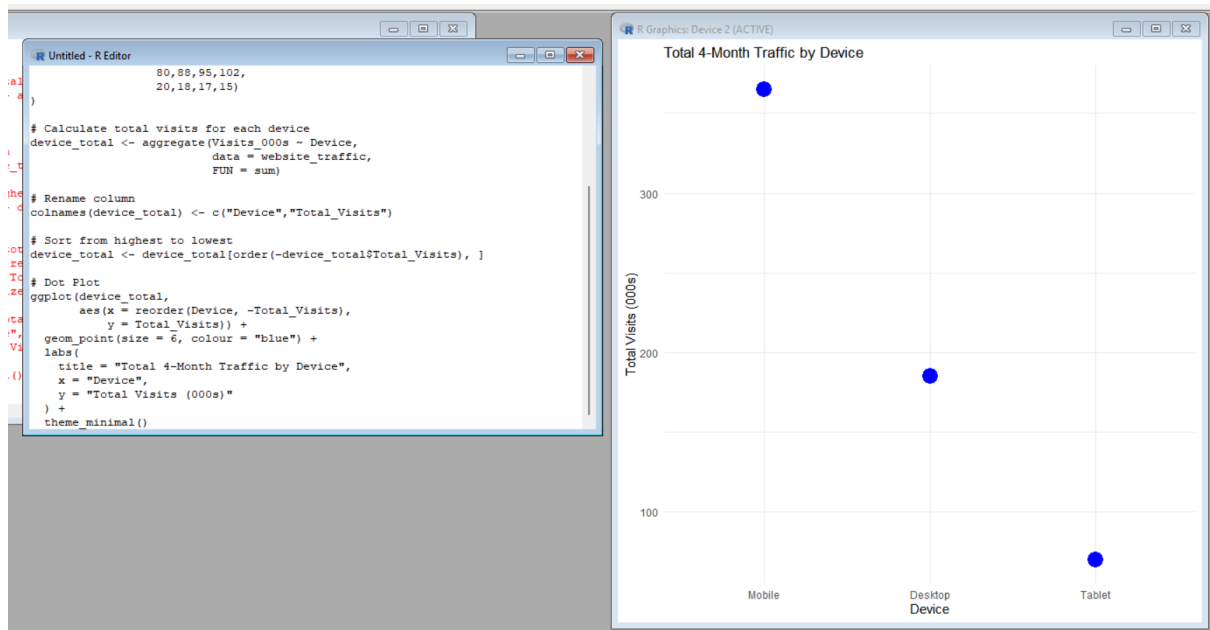
```
colnames(device_total) <- c("Device","Total_Visits")
```

```
# Sort from highest to lowest
```

```
device_total <- device_total[order(-  
device_total$Total_Visits), ]
```

```
# Dot Plot  
ggplot(device_total,  
        aes(x = reorder(Device, -Total_Visits),  
            y = Total_Visits)) +  
geom_point(size = 6, colour = "blue") +  
labs(  
    title = "Total 4-Month Traffic by Device",  
    x = "Device",  
    y = "Total Visits (000s)"  
) +  
theme_minimal()
```

Output:



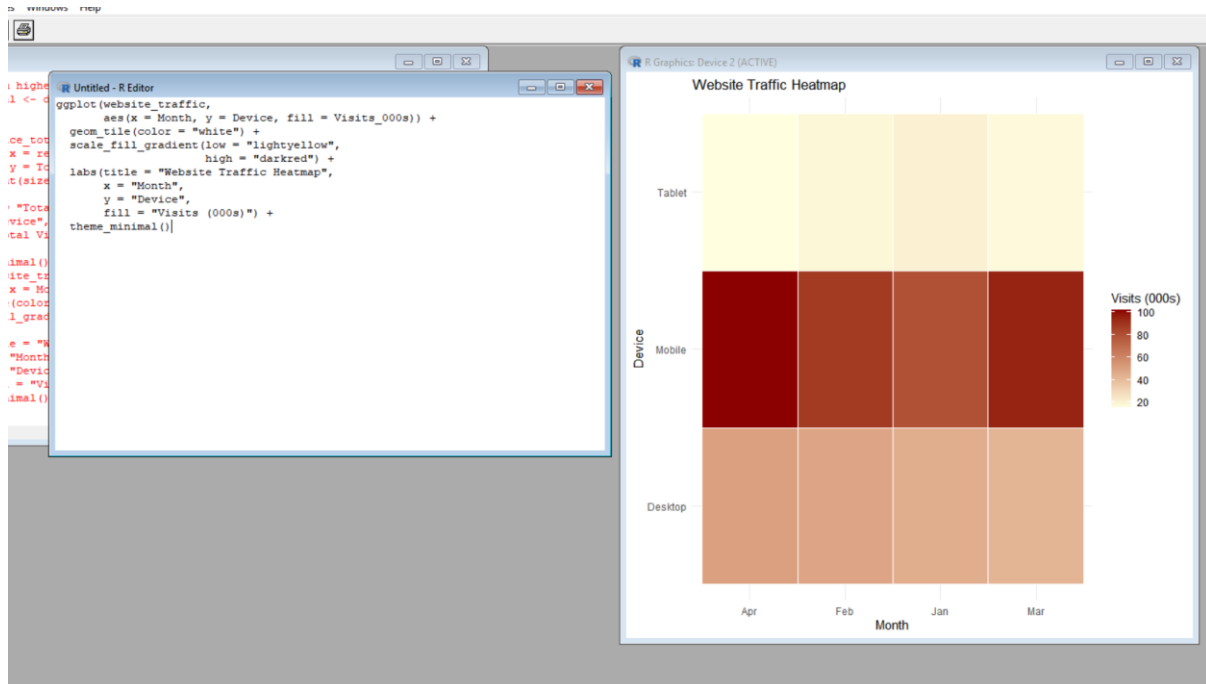
iv) R code for heatmap:

```

ggplot(website_traffic,
       aes(x = Month, y = Device, fill = Visits_000s)) +
  geom_tile(color = "white") +
  scale_fill_gradient(low = "lightyellow",
                     high = "darkred") +
  labs(title = "Website Traffic Heatmap",
       x = "Month",
       y = "Device",
       fill = "Visits (000s)") +
  theme_minimal()

```

Output:



Justification

i. Grouped Bar Chart

Justification:

A **grouped bar chart** is used to compare website traffic across different devices for each month. It allows easy comparison of Desktop, Mobile, and Tablet visits side by side within the same month.

ii. Stacked Bar Chart

Justification:

A **stacked bar chart** is used to show the total monthly website traffic while also displaying the contribution of

each device. It helps compare both the overall traffic and the individual device shares in each month.

iii. Dot Plot

Justification:

A **dot plot** is used to compare the total 4-month traffic for each device in a simple and uncluttered way. It makes it easy to rank devices from the highest to the lowest total traffic.

iv. Heatmap

Justification:

A **heatmap** is used to visualize website traffic across devices and months using color intensity. Darker colors represent higher visit counts, making it easy to identify patterns, trends, and high-traffic periods at a glance.

Q4) i) R code for violin plot:

```
# Install if needed
```

```
# install.packages("ggplot2")
```

```
library(ggplot2)
```

```
# Set seed
```

```
set.seed(7)
```

```
# Generate data
```

```
Route <- rep(c("Route A", "Route B", "Route C",  
"Route D"), each = 50)
```

```
Delivery_Time <- c(  
  rnorm(50, mean = 32, sd = 6),  
  rnorm(50, mean = 28, sd = 4),  
  rnorm(50, mean = 40, sd = 9),  
  rnorm(50, mean = 35, sd = 5)  
)
```

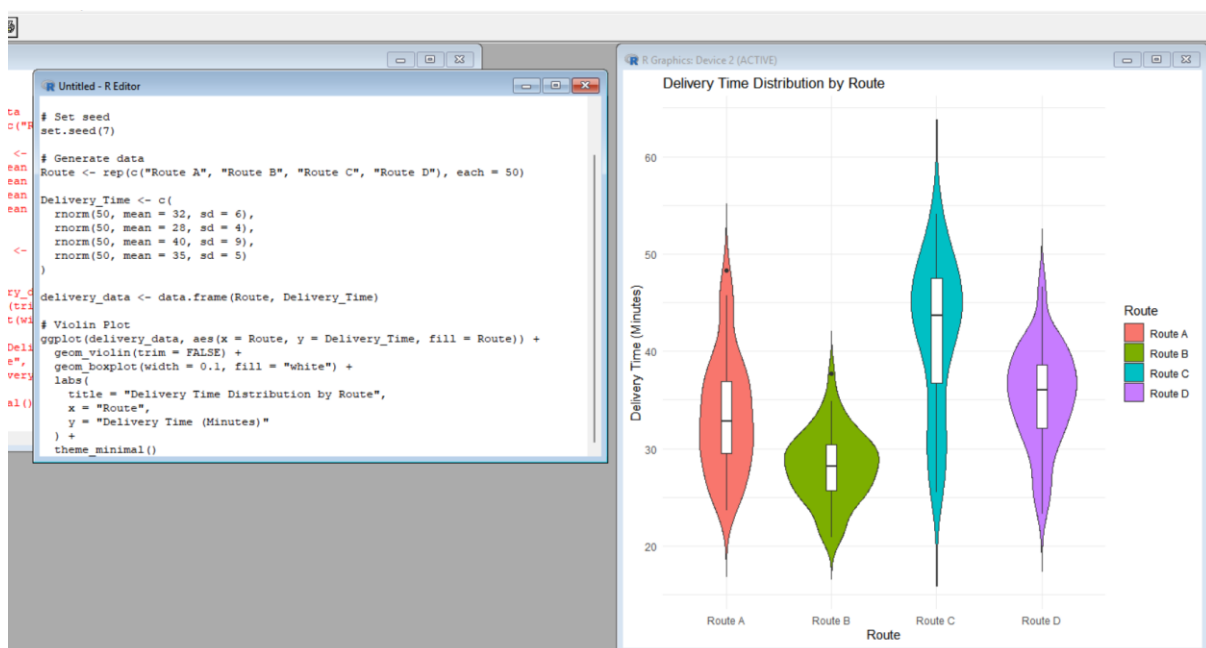
```
delivery_data <- data.frame(Route, Delivery_Time)
```

```
# Violin Plot
```

```
ggplot(delivery_data, aes(x = Route, y =  
Delivery_Time, fill = Route)) +  
  geom_violin(trim = FALSE) +  
  geom_boxplot(width = 0.1, fill = "white") +
```

```
labs(
  title = "Delivery Time Distribution by Route",
  x = "Route",
  y = "Delivery Time (Minutes)"
) +
theme_minimal()
```

Output:



ii) R code for ridgeline plot:

```
# Install if needed
```

```
# install.packages("ggridges")
```

```
library(ggplot2)
```

```
library(ggribes)
```

```
# Calculate mean delivery time
```

```
route_mean <- aggregate(Delivery_Time ~ Route,  
                        data = delivery_data,  
                        mean)
```

```
# Order routes by mean (fastest to slowest)
```

```
delivery_data$Route <- factor(  
  delivery_data$Route,  
  levels =  
  route_mean$Route[order(route_mean$Delivery_Time)]  
)
```

```
# Ridgeline Plot
```

```
ggplot(delivery_data,  
      aes(x = Delivery_Time,  
          y = Route,  
          fill = Route)) +
```

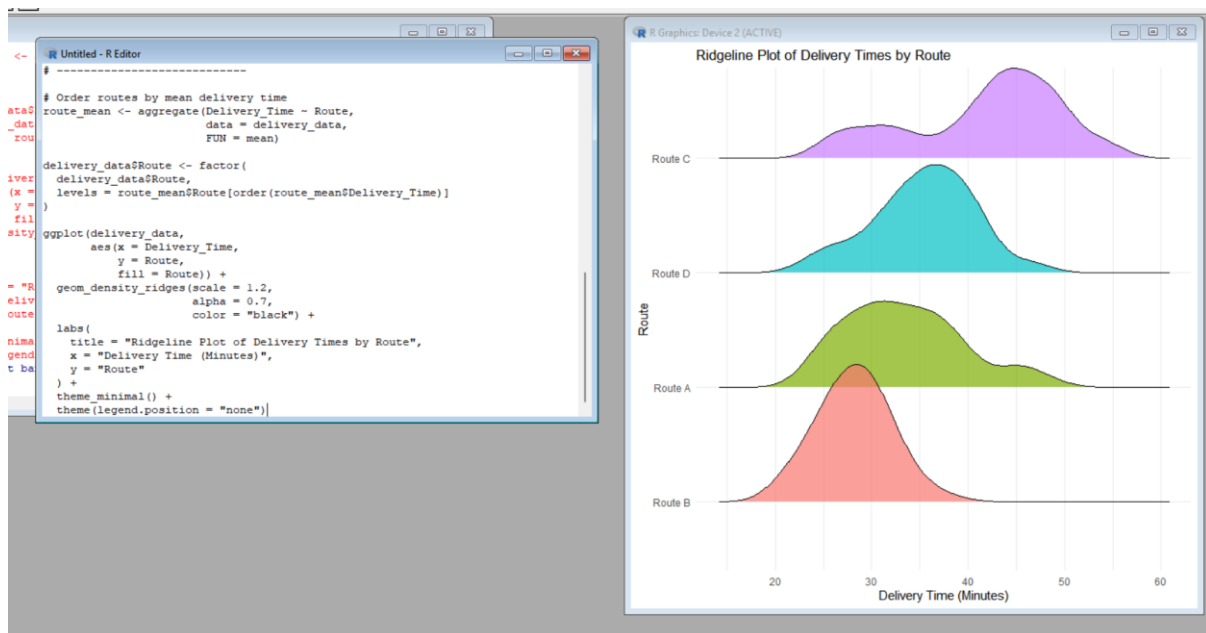


```

geom_density_ridges(alpha = 0.7) +
labs(
  title = "Ridgeline Plot of Delivery Times",
  x = "Delivery Time (Minutes)",
  y = "Route"
) +
theme_minimal() +
theme(legend.position = "none")

```

Output:



Justification:

11.violin plot

A violin plot is used to display the distribution of delivery times for each route. It combines the features of a box plot and a density plot, making it easy to compare the spread, median, and shape of the data across all four routes.

12. Ridgeline Plot

A ridgeline plot is used to compare the distribution of delivery times across multiple routes using overlapping density curves. Ordering the routes from the fastest to the slowest based on mean delivery time makes it easier to identify differences in delivery performance and variability.

Q5) R code histogram and density plot

```
# Load library
```

```
library(ggplot2)
```

```
# Load built-in dataset
```

```
data(iris)
```

```
ggplot(iris, aes(x = Petal.Length)) +
```

```
  geom_histogram(aes(y = after_stat(density)),
```

```
    bins = 20,  
    fill = "skyblue",  
    color = "black") +  
geom_density(color = "red", linewidth = 1.2) +  
labs(  
  title = "Histogram with Density Curve of Petal  
Length",  
  x = "Petal Length",  
  y = "Density"  
) +  
theme_minimal()  
ggplot(iris, aes(x = Petal.Length, fill = Species)) +  
  geom_histogram(aes(y = after_stat(density)),  
    bins = 20,  
    alpha = 0.5,  
    position = "identity",  
    color = "black") +  
  geom_density(aes(color = Species),  
    linewidth = 1) +  
labs(  

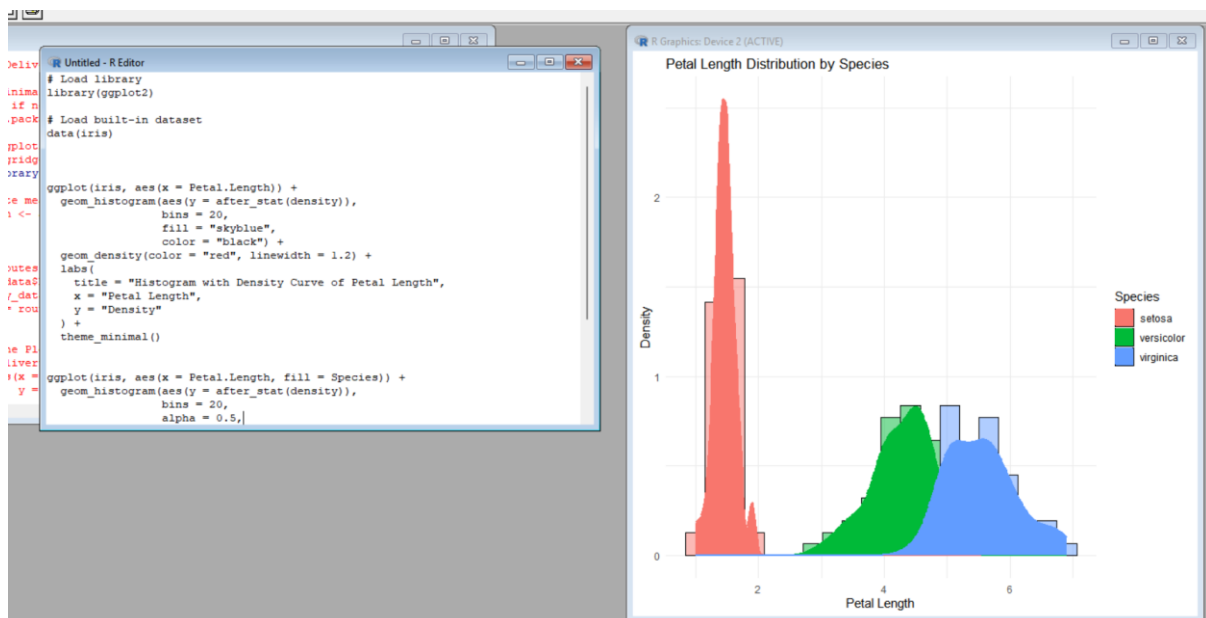
```

```

title = "Petal Length Distribution by Species",
x = "Petal Length",
y = "Density"
) +
theme_minimal()

```

Output:



i) Histogram

Justification:

A **histogram** is used to display the frequency distribution of a continuous variable. It helps visualize the shape, spread, and concentration of the Petal Length values.

ii) Density Plot (KDE)

Justification:

A **density plot** provides a smooth estimate of the data distribution. Overlaying it on the histogram makes it easier to identify the overall distribution pattern and compare it with the histogram.